

JavaScript Functions Guide

Function Declaration

```
function add(a, b) {  
  return a + b;  
}  
console.log(add(2,3));
```

Function Expression

```
const add = function(a, b) {  
  return a + b;  
};  
console.log(add(2,3));
```

Arrow Function

```
const add = (a, b) => a + b;  
console.log(add(2,3));
```

Default Parameters

```
function greet(name = 'Guest') {  
  return name;  
}  
console.log(greet());
```

Rest Parameters

```
function sum(...nums) {  
  let total = 0;  
  for(let i=0;i<nums.length;i++){  
    total += nums[i];  
  }  
  return total;  
}  
console.log(sum(1,2,3));
```

Callback Function

```
function process(name, cb){  
  cb(name);  
}  
process('Ali', function(n){ console.log(n); });
```

IIFE

```
(function(){  
  console.log('Run');  
})();
```

Recursion

```
function factorial(n){  
  if(n===1) return 1;  
  return n * factorial(n-1);  
}  
console.log(factorial(5));
```